

David Xu

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Presentation

I am currently a Research Fellow (postdoc) at Korea Institute for Advanced Study (KIAS) in Seoul, South Korea. I am attached to the School of Mathematics and my mentor is Sang-hyun Kim. I defended my Ph.D. thesis on July 12th 2024.

My research interests revolve around hyperbolic geometry, group actions and analytic properties of groups. I work mainly on the *infinite-dimensional hyperbolic space* and *representations into its groups of isometries* with various *discreteness properties*. I also started recently to wonder about infinite-dimensional higher rank symmetric spaces in general.

Employment

- 2025–current **Research Fellow (postdoc)**, *Korea Institute for Advanced Study (KIAS)*, Seoul.
- 2023–2024 **ATER (teaching and research assistant)**, *IUT de Sceaux*.

Education

- 2020–2024 **Ph.D. in Mathematics**, *Université de Lorraine*, Nancy, Discrete isometry groups of the infinite-dimensional hyperbolic space.
Supervised by Bruno Duchesne and Gye-Seon Lee
- 2019–2020 **Master degree in Mathematics**, *Université de Strasbourg*, Strasbourg.
- 2018–2019 **Agrégation externe de Mathématiques, option D Informatique**, *ENS Rennes and Université de Rennes 1*, Rennes, rank : 110.
- 2016–2020 **Magistère de Mathématiques**, *ENS Rennes*.
- 2016–2018 **Bachelor degree in Computer Science**, *Université de Rennes 1*.
- 2016–2017 **Bachelor degree in Mathematics**, *Université de Rennes 1*.

Publications and preprints

- 2025 **An explicit exotic representation of a rank-one simple Lie group via convex bodies**, joint work with François Fillastre et Yusen Long, Preprint, Arxiv link.
- 2024 (v2 in 2026) **Convex-cocompact representations into the isometry group of the infinite-dimensional hyperbolic space**, Preprint, Arxiv link.
- 2024 **Discrete isometry groups of the infinite-dimensional hyperbolic space**, *Ph.D. thesis*, HAL link.

Invited talks

- 11.27.2025 **Dynamical group theory VII**, *SNU*, Seoul.
- 10.22.2025 **Fall Meeting of the Korean Mathematical Society**, Seoul.
- 07.29.2025 **Differential Topology Seminar**, *Kyoto University*, Kyoto.

06.12.2025 **"2K-GATE Workshop I"**, *KIAS*, Seoul.

04.25.2025 **Spring Meeting of the Korean Mathematical Society**, *KAIST*, Daejeon.

02.19.2025 **VISGAT seminar**, *KIAS*, Seoul.

01.21.2025 **SNU Exchange Seminar**, *SNU*, Seoul.

12.09.2024 **Conference "Arithmetic groups, hyperbolic manifolds and computation"**, Bordeaux.

11.26.2024 **Geometry Seminar**, *MPI MiS*, Leipzig.

10.10.2024 **Geometry and Topology Seminar**, *I2M*, Marseille.

05.27.2024 **Ergodic Theory Seminar**, *IRMAR*, Rennes.

04.18.2024 **Workshop "Submanifolds of locally symmetric spaces"**, *ANR GAPR*, Autrans.

02.13.2024 **Online Seminar of Geometry and Topology**, *Paris-Saclay/SNU*.

06.06.2023 **Ph.D. Students Seminar Pampers**, *IRMAR*, Rennes.

05.03.2023 **Ph.D. Students Seminar**, *IECL*, Nancy.

01.30.2023 **Groups and Operator Algebras Seminar**, *IMO*, Orsay.

05.19.2022 **Ph.D. Students Seminar**, *IRMA*, Strasbourg.

05.12.2022 **Ph.D. Students Seminar**, *Institut Fourier*, Grenoble.

09.23.2021 **"Groups and Geometry" meeting at maison du Kleebach**, *ANR AGIRA*, Kleebach.

29/04/2021 **Ph.D. Students Seminar**, *IRMA*, Strasbourg.

Conferences organized

June 2026 **Workshop on infinite-dimensional geometry**, *MPI MiS*, Leipzig.
(upcoming) With Clarence Kineider and Eugen Rogozinnikov

Oct. 2025 **Mapping Class Groups at KIAS**, *KIAS*, Seoul.
With Carl-Fredrik Nyberg Brodda, Sang-hyun Kim, Sanghoon Kwak and Eugen Rogozinnikov

Teaching

2023–2025 **Numerical tools for management**, *IUT de Sceaux*, Sceaux, for 2nd year undergraduate students in management and marketing.
Lectures and classes

2023–2025 **Mathematical tool for management**, *IUT de Sceaux*, Sceaux, for 1st year undergraduate students management and marketing.
Lectures and classes

2022–2023 **Mathematics for engineers**, *Université de Lorraine*, Nancy, for 1st year undergraduate students in engineering curriculum.
Lectures and classes

2021–2022 **Analysis**, *Université de Lorraine*, Nancy, for 2nd year undergraduate students in mathematics.
Classes

2021–2022 **Financial mathematics**, *IUT Nancy-Charlemagne*, Nancy, for 1st year undergraduate students in management.
Tutorials

2021–2022 **Numerical analysis**, *ENSEM*, Nancy, for 3rd year undergraduate students in engineering school.
Classes and practical work (Matlab)

2020–2022 **Numerical analysis**, *Université de Lorraine*, Nancy, for 2nd year undergraduate students in mathematics.
Classes and practical work (Matlab)

2020–2021 **Mathematical tool for management**, *IUT Nancy-Charlemagne*, Nancy, for 1st year undergraduate students in management and marketing.
Classes

Grants, awards and honors

2020–2021 **LUE Graduate**, *Lorraine Université d'Excellence*, Nancy.
Grant from the *DrEAM* project for funding a two months visit at *Seoul National University*, Seoul, South Korea

Visits and internships

September 2025 **Research visit at Institut Fourier**, Grenoble.
Guest of Anne Parreau and Balthazar Fléchelles

July 2025 **Research visit at Kyoto University**, Kyoto.
Guest of Koji Fujiwara and Alexis Marchand

May 2025 **Research visit at MPI MiS**, *Max Planck Institute for Mathematics in the Sciences*, Leipzig, Germany.
Guest of Anna Wienhard and Clarence Kineider

October 2024 **Research visit at SNU**, *Seoul National University*, Seoul, South Korea.
Visit funded by the PHC-STAR program, guest of Gye-Seon Lee

August 2024 **Research visit at EPFL**, *École Polytechnique Fédérale de Lausanne*, Lausanne, Switzerland.
Guest of Gonzalo Ruiz Stolorowicz

March–April 2022 **Visiting student**, *Seoul National University*, Seoul, South Korea.
Visit of Seoul National University funded by the DrEAM project from LUE "Lorraine Université d'Excellence", supervised by Gye-Seon Lee

May–July 2020 **Master internship**, *Institut Elie Cartan*, Nancy, France.
Master internship on the "isometries of the infinite-dimensional hyperbolic space", supervised by Bruno Duchesne

May–July 2018 **Master internship**, *Mathematisches Institut der Universität Heidelberg*, Heidelberg, Germany.
Master internship on "hyperbolic geometry", supervised by Anna Wienhard and Gye-Seon Lee

May–June 2017 **Bachelor internship**, *Institut Camille Jordan*, Lyon, France.
Bachelor internship on "valued fields in model theory", supervised by Itai Ben Yaacov

May 2013 **Highschool internship**, *Institut Camille Jordan*, Lyon, France.
Highschool "discovery" internship organised by the association *Science Académie* on differential equations in compartmental models (SIR model)

Some other experiences, responsibilities and outreach activities

March 2026 **π -day at KIAS**, *KIAS*, Seoul.
Organization of a mathematical event for elementary school students

August 2025 **과학 아카데미 (« Science Academy »)**, *KIAS*, Seoul.
Organization of a mathematical event for elementary school students

March 2025 **π -day at KIAS**, *KIAS*, Seoul.
Organization of a mathematical event for elementary and middle school students

2025–current **VISGAT Seminar**, *KIAS/KAIST*.
Organizer of the VISGAT Seminar

2021–2023 **MATH.en.JEANS**, *Collège Edmond de Goncourt*, Pulnoy.
Animating *MATH.en.JEANS* workshops for middle school students

2020–2022 **Doctoral students' representative at the IAEM doctoral school**, *Université de Lorraine*, Nancy.

- June 2017 **Exam preparation tutor**, *Lycée La Martinière Monplaisir*, Lyon.
Tutoring for the oral competitive exams of CPGE
- 2016 **Mathematics tutor**, *Lycée Émile Zola*, Rennes.
Tutoring for highschool students
- March 2013 **Music teacher**, *École de musique Guy Candeloro*, Lyon.
Replacement of a music theory teacher and an accordion teacher in a music school for 1 month